



MTH643 - INTRODUCTION TO MATLAB

From: ABU SULTAN

Page 14 of 22

Important and Solved Questions for Mid Term Exam



```

clc
clear all
close all
eq1='Dx=x+2*y-z';
eq2='Dy=x+z';
eq3='Dz=4*x-4*y+5*z';
incond='x(0)=1,y(0)=2,z(0)=3';
[x,y,z]=dsolve(eq1,eq2,eq3,incond)
t=linspace(0,0.5,25);
xx=eval(vectorize(x));
yy=eval(vectorize(y));
zz=eval(vectorize(z));
plot(t,xx,t,yy,t,zz)
xlabel('x-axis')
ylabel('y-axis')
title('Graph of solution curves of DE')
grid

```

L07_Q37: Solve the following system of differential equations.

$$x''(t) = x'(t) + 2y'(t) - z'(t)$$

$$y''(t) = x'(t) + z'(t)$$

$$z''(t) = 4x'(t) - 4y'(t) + 5z'(t)$$

Corresponding to the following conditions $x(0) = 1, x'(0) = 2, y(0) = 1, y'(0) = 2, z(0) = 3, z'(0) = 4$.

Moreover plot the solution curves using the plot command.

(P#210-213 - EXERCISE - LECT#07)

%Solve the following system of differential equations.

%x''(t)=x'(t)+2y'(t)-z'(t)

%y''(t)=x'(t)+z'(t)

%z''(t)=4x'(t)-4y'(t)+5z'(t)

%Corresponding to the following conditions x(0)=1,x'(0)=2,y(0)=1,y'(0)=2,z(0)=3,z'(0)=4.

%Moreover plot the solution curves using the plot command.

```

clc
clear all
close all
eq1='D2x=Dx+2*y-Dz';
eq2='D2y=Dx+Dz';
eq3='D2z=4*Dx-4*y+5*z';
incond='x(0)=1,Dx(0)=2,y(0)=1,Dy(0)=2,z(0)=3,Dz(0)=4';
[x,y,z]=dsolve(eq1,eq2,eq3,incond)
t=linspace(0,0.5,25);
xx=eval(vectorize(x));
yy=eval(vectorize(y));
zz=eval(vectorize(z));
plot(t,xx,t,yy,t,zz)
xlabel('x-axis')
ylabel('y-axis')
title('Graph of solution curves of DE')
grid

```



MTH643 - INTRODUCTION TO MATLAB

From: ABU SULTAN

Page 15 of 22

Important and Solved Questions for Mid Term Exam



L07_Q38: Solve the following system of differential equations.

$$\dot{x}(t) = 2x'(t) + 2y'(t) - z'(t)$$

$$\dot{y}(t) = 3x'(t) - 3z'(t)$$

$$\dot{z}(t) = 5x'(t) - 4y'(t) + 5z'(t)$$

Corresponding to the following conditions $x(0) = 2, x'(0) = 1, y(0) = 3, y'(0) = 4, z(0) = 3, z'(0) = 5$ (P#216 - EXERCISE - LECT#07)

%Solve the following system of differential equations.

%x''(t)=2x'(t)+2y'(t)-z'(t)

%y''(t)=3x'(t)-3z'(t)

%z''(t)=5x'(t)-4y'(t)+5z'(t)

%Corresponding to the following conditions x(0)=2,x'(0)=1,y(0)=3,y'(0)=4,z(0)=3,z'(0)=5

clc

clear all

close all

eq1='D2x=2*Dx+2*Dy-Dz'

eq2='D2y=3*Dx-3*Dz';

eq3='D2z=5*Dx-4*Dy+5*Dz';

incond='x(0)=2,Dx(0)=1,y(0)=3,Dy(0)=4,z(0)=3,Dz(0)=5';

[x,y,z]=dsolve(eq1,eq2,eq3,incond)

t=linspace(0,5,25);

xx=eval(vectorize(x));

yy=eval(vectorize(y));

zz=eval(vectorize(z));

plot(t,xx,t,yy,t,zz)

xlabel('x-axis')

ylabel('y-axis')

title('Graph of solution curves of DE')

grid

L07_Q39: Solve the following system of differential equations.

$$\dot{x}(t) = 2x(t) + 2y(t) - 3z(t)$$

$$\dot{y}(t) = 2x(t) + 3z(t)$$

$$\dot{z}(t) = x(t) - y(t) + z(t)$$

Corresponding to the following conditions $x(0)=2, y(0)=1, z(0)=4$. Moreover plot the solution curves using plot command. (P#217 - EXERCISE - LECT#07)

%Solve the following system of differential equations.

%x'(t)=2x(t)+2y(t)-3z(t)

%y'(t)=2x(t)+3z(t)

%z'(t)=x(t)-y(t)+z(t)

%Corresponding to the following conditions x(0)=2,y(0)=1,z(0)=4.

%Moreover plot the solution curves using plot command.

clc

clear all

close all

eq1='Dx=2*x+2*y-3*z'

eq2='Dy=2*x+3*z';

eq3='Dz=x-y+z';

incond='x(0)=2,y(0)=1,z(0)=4';

[x,y,z]=dsolve(eq1,eq2,eq3,incond)

t=linspace(0,5,25);

xx=eval(vectorize(x));

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**MTH643 - INTRODUCTION TO MATLAB**

From: ABU SULTAN

Page 9 of 22

**Important and Solved Questions for Mid Term Exam**

L05_Q23: Differentiate the following functions using diff command. Moreover plot the multiple graphs of the given functions and their derivatives on the interval [0, 5]. $f(x) = x^2 + 2x + 10$ and $g(x) = \sin(x)$ (P#116 – LECT#05)

%Differentiate the following functions using diff command.

%Moreover plot the multiple graphs of the given functions and their derivatives on the interval [0, 5].

% $f(x) = x^2 + 2x + 10$ and $g(x) = \sin(x)$

clc

clear all

close all

syms x

$f = x^2 + 2x + 10$

$g = \sin(x)$

$f1 = \text{diff}(f)$

$g1 = \text{diff}(g)$

$x = 0:0.1:5;$

$\text{plot}(x, (x.^2 + 2.*x + 10), x, (2.*x + 2), x, \sin(x), x, \cos(x))$

$\text{xlabel}('x\text{-axis}')$

$\text{ylabel}('y\text{-axis}')$

$\text{title}('Graph of function and its derivatives using diff and plot')$

grid

L06_Q24: Compute the values of $\sin(x)$ at $x = 0; 1$, and 2 using for loop.

(P#170 – LECT#06)

%Compute the values of $\sin(x)$ at $x = 0; 1$, and 2 using for loop.

clear all

close all

clc

for $i = 1:5$

$y = \sin(i)$

end

L06_Q25: Solve the following algebraic equations using Script M-Files.

$$x^2 + 2x - 4 = 0 \text{ and } x^3 + 5x^2 + 4x + 3 = 0$$

Solve the following equations for x using Script M-Files.

$$x + \log(y) = 3 \text{ and } x + 10y = 6$$

Solve the following system of equations using Script M-Files.

$$x + y^2 = 2, y - 3x = 7 \text{ and } x + y^2 = 2, 2y^2 - 3x = 7$$

(P#173 - EXERCISE – LECT#06)

%Solve the following algebraic equations using Script M-Files.

% $x^2 + 2x - 4 = 0$ and $x^3 + 5x^2 + 4x + 3 = 0$

%Solve the following equations for x using Script M-Files. $x + \log(y) = 3$ and $x + 10y = 6$

%Solve the following system of equations using Script M-Files.

% $x + y^2 = 2, y - 3x = 7$ and $x + y^2 = 2, 2y^2 - 3x = 7$

clear all

close all

clc

syms x y

$[x] = \text{solve}('x^2 + 2.*x - 4 = 0', 'x^3 + 5.*x^2 + 4.*x + 3 = 0')$

$[x, y] = \text{solve}('x + \log(y) = 3', 'x + 10.*y = 6');$

$[x, y] = \text{solve}('x + y.^2 = 2', 'y - 3.*x = 7');$

$[x, y] = \text{solve}('x + y.^2 = 2', '2.*y.^2 - 3.*x = 7');$



MTH643 - INTRODUCTION TO MATLAB

From: ABU SULTAN

Page 11 of 22

Important and Solved Questions for Mid Term Exam



```
syms x y
[X,Y]=meshgrid(-2*pi:0.01:2*pi,0:0.01:4*pi);
contour(X,Y,sin(X)+cos(Y))
xlabel('x-axis');
ylabel('y-axis');
title('Contour Plot of Sin(X)+Cos(Y)')
grid
```

L07_Q29: Find the solution of the following initial value problem $dy/dx = x^2 + y$; $y(1) = 3$. Moreover, plot the graph of the solution using `ezplot` command on the interval $[0,2]$. (P#187-189 - EXERCISE - LECT#07)

```
%Find the solution of the following initial value problem dy/dx = x^2 + y; y(1) = 3.
%Moreover, plot the graph of the solution using ezplot command on the interval [0,2].
```

```
clc
clear all
close all
syms y
eq1='Dy=x^2+y';
y=dsolve(eq1,'y(1)=3','x')
ezplot(y,[0,2])
xlabel('x-axis')
ylabel('y-axis')
title('Solution of Initial Value Problem and ezplot of solution')
grid
```

L07_Q30: Find general solution of the following boundary value problem $y''(x) + 8y'(x) + 2y(x) = \cos(x)$. Then plot the solution curve on the interval $[0,4]$ corresponding to the following boundary conditions. $y(0) = 4, y'(1) = 5$. (P#215 - EXERCISE - LECT#07)

```
%Find general solution of the following boundary value problem y''(x)+8y'(x)+2y(x)=cos(x).
%Then plot the solution curve on the interval [0,4]
%corresponding to the following boundary conditions. y(0)=4,y'(1)=5.
```

```
clc
clear all
close all
%syms x y
eq1='D2y+8*Dy+2*y=cos(x)';
icon='y(0)=4,Dy(1)=5';
y=dsolve(eq1,icon,'x')
ezplot(y,[0,4])
xlabel('x-axis')
ylabel('y-axis')
title('Solution of Boundary Value Problem and Solution Curve')
grid
```

L07_Q31: Plot the family of curves of the following differential equation on the interval $[0,3]$, with initial values $y(0) = -3 : 3$, $dy/dx = x * y + 2 * x^2$. (P#194-196 - EXERCISE - LECT#07)

```
%Plot the family of curves of the following differential equation on the interval [0,3].
%with initial values y(0) = -3 : 3, dy/dx = x * y + 2 * x^2.
```

```
clc
clear all
```



MTH643 - INTRODUCTION TO MATLAB

From: ABU SULTAN

Page 8 of 22

Important and Solved Questions for Mid Term Exam



%Moreover plot the multiple graph of the function and their derivatives by defining X and Y using meshgrid command.

%Then define Z and create a surface plot:

%function is $f(x,t)=xe^{(-x^2-t^2)}$ x,t?[-2,2]

```
clc
clear all
close all
syms x t
f=x*exp(-x^2-t^2);
f1=diff(f,x)
f2=diff(f,t)
[x,t]=meshgrid(-2:0.2:2,-2:0.2:2);
ff=eval(vectorize(f));
f11=eval(vectorize(f1));
f22=eval(vectorize(f2));
surf(x,t,ff)
hold on
surf(x,t,f11)
hold on
surf(x,t,f22)
hold off
```

L05_Q21: Find the first-order, second-order, and fourth-order derivatives of the function $f(x) = 2xy + (xy)^3$ with respect to variable y using diff command.

%Find the first-order, second-order, and fourth-order derivatives of the function

% $f(x)=2xy+(xy)^3$ with respect to variable y using diff command.

```
syms x y
f=2*x*y+(x*y)^3
diff(f,y,1)
diff(f,y,2)
diff(f,y,4)
```

L05_Q22: Find the integral of the function $f(x) = x^2 + 10x + 4$ using int command. Plot the multiple graphs of the function and its output using plot command. Label the horizontal axis with x and vertical axis with y. Keeps the title "Graph of function and its integral"

%Find the integral of the function $f(x)=x^2+10x+4$ using int command.

%Plot the multiple graphs of the function and its output using plot command.

%Label the horizontal axis with x and vertical axis with y.

%Keeps the title "Graph of function and its integral"

```
clc
clear all
close all
syms x
f=int(x^2+10*x+4)
x=-2:0.01:2;
plot(x,x.^2+10.*x+4,(x.*(x.^2 + 15.*x + 12))./3,x)
xlabel('x-axis')
ylabel('y-axis')
title('Graph of function and its integral')
grid
```



MTH643 - INTRODUCTION TO MATLAB

From: ABU SULTAN

Page 10 of 22

Important and Solved Questions for Mid Term Exam



MIX_Q26: Find the first and second derivative of the following function with respect to x using MATLAB command. Then create a surface plot of the given function and its first and second derivatives by defining X and Y using mesh grid command with hold on and hold off command. $f(x, y) = xy(x^2 - y^2)$. (ASSIGNMENT#01)

% ASSIGNMENT#1 BY ME

% FIND FIRST & SECOND DERIVATIVES OF $f(x,y)=xy(x^2-y^2)$ w.r.t x .

% Then create surface plot of given function, its first & second derivatives using

% meshgrid command with hold on and hold off command.

```
clear all;
clc;
close all;
syms x y;
f=x*y*(x^2-y^2)
df=diff(f)
ddf=diff(df)
u=-2:0.1:2;
v=-2:0.1:2;
[x,y]=meshgrid(u,v);
f=eval(vectorize(f));
df=eval(vectorize(df));
ddf=eval(vectorize(ddf));
surf(x,y,f)
hold on
surf(x,y,df)
hold on
surf(x,y,ddf)
hold off
```

MIX_Q27: Solve: $f(x, y) = xy^2 + 2x + 4 = 0$, $f(x, y) = 3x^2y + 2xy^2 + 4x$ at $x=5$,

$f(x) = \sin x$ at $x = 5$

% Solve: $f(x,y)=xy^2+2x+4=0$, $f(x,y)=3x^2y+2xy^2+4x$ at $x=5$

% $f(x)=\sin x$ at $x=5$

```
clc
clear all
close all
syms x y u v
[x,y]=solve('x*y^2+2*x+4=0')
u=3*x^2*y+2*x*y^2+4*x;
subs(u,x,5)
v=sin(x);
subs(v,x,5)
```

MIX_Q28: Plot the contour plots of the $\sin(x) + \cos(y)$ where x range from -2π to 2π , y range from 0 to 4π . Label the horizontal axis with x and vertical axis with y . Keep the title of the graph "Contour Plot of $\sin(X) + \cos(Y)$."

% Plot the contour plots of the $\sin(x) + \cos(y)$

% where x range from -2π to 2π , y range from 0 to 4π .

% Label the horizontal axis with x and vertical axis with y .

% Keep the title of the graph "Contour Plot of $\sin(X) + \cos(Y)$."

```
clc
clear all
close all
```



MTH643 - INTRODUCTION TO MATLAB

From: ABU SULTAN

Page 4 of 22



Important and Solved Questions for Mid Term Exam

L04_Q12: Plot the graph of $x^3 + x^2 + 4$ on the interval $-2, 2]$ using `ezplot`, hold on and hold off commands. Label the horizontal axis with x and vertical axis with y . Finally gives the title " $x^3 + x^2 + 4$ ". (Follow P#90-92 - LECT#04)

%Plot the graph of x^3+x^2+4 on the interval $-2, 2]$ using `ezplot`, hold on and hold off commands.

%Label the horizontal axis with x and vertical axis with y .

%Finally gives the title " x^3+x^2+4 ".

`clc`

`clear all`

`close all`

`syms x`

`x=x^3+x^2+4`

`ezplot(x,-2,2)`

`axis square`

`xlabel('x')`

`ylabel('y')`

`title('x^3+x^2+4')`

L04_Q13: Plot the graph of the function $\sin(x) + \cos(x)$ on the interval $[0; 1]$, using plot command. Label the horizontal axis with x and vertical axis with y . Make the shape of the graph square. Keeps the title of the graph "The Graph of $\sin(x) + \cos(x)$ ". Keeps the layout of the graph as grid. (Follow P#98-100 - LECT#04)

%Plot the graph of the function $\sin(x) + \cos(x)$ on the interval $[0; 1]$, using plot command.

%Label the horizontal axis with x and vertical axis with y .

%Make the shape of the graph square.

%Keeps the title of the graph "The Graph of $\sin(x) + \cos(x)$ ".

%Keeps the layout of the graph as grid.

`clc`

`clear all`

`close all`

`X=0:0.01:1;`

`plot(X,(sin(X)+cos(X)))`

`xlabel('x')`

`ylabel('y')`

`axis square;`

`title('The Graph of  $\sin(x) + \cos(x)$ ');`

`grid`

L04_Q14: Plot the multiple curves of the functions $\sin(x) + \cos(x)$ and $e^{-x} + x^2$ on the interval $[0; 4]$, using `ezplot` command. Label the horizontal axis with x and vertical axis with y . Make the shape of the graph square. Keeps the title of the graph "The Graph of Multiple curves using `ezplot` command ". Keeps the layout of the graph as grid. (Follow P#90-92 - LECT#04)

%Plot the multiple curves of the functions $\sin(x) + \cos(x)$ and $e^{-x} + x^2$

%on the interval $[0; 4]$, using `ezplot` command.

%Label the horizontal axis with x and vertical axis with y .

%Make the shape of the graph square.

%Keeps the title of the graph "The Graph of Multiple curves using `ezplot` command ".

%Keeps the layout of the graph as grid.

`clc`

`clear all`

`close all`



MTH643 - INTRODUCTION TO MATLAB

From: ABU SULTAN

Page 7 of 22



Important and Solved Questions for Mid Term Exam

```
f=log(x)+sin(x)
f1=diff(f,1)
f4=diff(f,4)
f5=diff(f,5)
x=0:0.1:1;
f11=eval(vectorize(f1));
f44=eval(vectorize(f4));
f55=eval(vectorize(f5));
plot(x,f11,x,f44,x,f55)
xlabel('x-axis')
ylabel('y-axis')
title('Multiple plot of the function and its derivatives')
grid
```

L05_Q19: Find the first-order, fourth-order, and fifth-order derivatives of the function $f(x) = \ln x * y + \sin(x^2)$ using diff command. Plot the multiple graphs of the outputs using plot command on the interval [0, 1] and without using hold on and hold off commands Label the horizontal axis with x and vertical axis with y. Keeps the title "Multiple plot of the function and its derivatives". All the coding should be written using Script M-File. (P#145 - EXERCISE - LECT#05) + (P#176 - EXERCISE - LECT#06)

```
%Find the first-order, fourth-order, and fifth-order derivatives of the function
% f(x)=ln x*y+sin(x^2) using diff command.
%Plot the multiple graphs of the outputs using plot command on the interval [0, 1]
%and without using hold on and hold off commands.
%Label the horizontal axis with x and vertical axis with y.
%Keeps the title "Multiple plot of the function and its derivatives".
%All the coding should be written using Script M-File
clc
clear all
close all
syms x y
f=log(x*y)+sin(x^2)
d1=diff(f,1)
d4=diff(f,4)
d5=diff(f,5)
x=0:0.1:1;
d11=eval(vectorize(d1));
d44=eval(vectorize(d4));
d55=eval(vectorize(d5));
plot(x,d11,x,d44,x,d55)
xlabel('x-axis')
ylabel('y-axis')
grid
title('Multiple plot of the function and its derivatives')
```

L05_Q20: Find the derivative of the following function with respect to x, and t variables using diff command. Moreover plot the multiple graph of the function and their derivatives by defining X and Y using meshgrid command, Then define Z and create a surface plot: $f(x, t) = xe^{-x^2-t^2}$ x, t $\in [-2, 2]$. (P#122 - LECT#05)

```
%Find the derivative of the following function with respect to x, and t variables using diff
command.
```




MTH643 - INTRODUCTION TO MATLAB

From: ABU SULTAN

Page 6 of 22

Important and Solved Questions for Mid Term Exam



```
ylabel('y')
title('The Graph of Multiple Curves using plot command')
grid
```

L05_Q17: Find the first-order, second-order, and fourth-order derivatives of the function $f(x) = x^{10} + 2 \cdot (x^2) + \sin(x)$ Using diff command. Plot the multiple graphs of the outputs using ezplot command on the interval $[-\pi, \pi]$. Label the horizontal axis with x and vertical axis with y. Keeps the title "Multiple plot of the function and its derivatives". All the coding should be written using Script M-File.

(P#144 - EXERCISE - LECT#05) + (P#174 - EXERCISE - LECT#06)

```
%Find the first-order, second-order, and fourth-order derivatives of the
function f(x)=x^10+2*(x^2)+sin(x) Using diff command.
%Plot the multiple graphs of the outputs using ezplot command on the interval [-pi,pi].
%Label the horizontal axis with x and vertical axis with y.
%Keeps the title "Multiple plot of the function and its derivatives".
%All the coding should be written using Script M-File
clc
clear all
close all
syms x
f=x^10+2*x^2+sin(x);
f1=diff(f,1)
f2=diff(f,2)
f4=diff(f,4)
ezplot(f,[-pi,pi])
hold on
ezplot(f1,[-pi,pi])
hold on
ezplot(f4,[-pi,pi])
hold off
xlabel('x-axis')
ylabel('y-axis')
title('Multiple plot of the function and its derivatives')
grid
```

L05_Q18: Find the first-order, fourth-order, and fifth-order derivatives of the function $f(x) = \ln x + \sin(x)$ using diff command. Plot the multiple graphs of the outputs using plot command on the interval $[0, 1]$ and without using hold on and hold off commands Label the horizontal axis with x and vertical axis with y. Keeps the title "Multiple plot of the function and its derivatives". All the coding should be written using Script M-File.

(P#145 - EXERCISE - LECT#05) + (P#175 - EXERCISE - LECT#06)

```
%Find the first-order, fourth-order, and fifth-order derivatives of the function
f(x)=ln x+sin(x) using diff command.
%Plot the multiple graphs of the outputs using plot command on the interval [0, 1]
%and without using hold on and hold off commands.
%Label the horizontal axis with x and vertical axis with y.
%Keeps the title "Multiple plot of the function and its derivatives".
%All the coding should be written using Script M-File
clc
close all
clear all
syms x
```



MTH643 - INTRODUCTION TO MATLAB

From: ABU SULTAN

Page 3 of 22



Important and Solved Questions for Mid Term Exam

L04_Q09: Plot the graph of e^{-x} and $\sin(x)$ on the interval $[0, 8]$ using plot, hold on and hold off commands. Label the horizontal axis with x and vertical axis with y. Finally gives the title "Multiple Graphs". (P#90-92 - LECT#04)

%Plot the graph of $e^{(-x)}$ and $\sin(x)$ on the interval $[0, 8]$ using plot, hold on and hold off.

%Label the horizontal axis with x and vertical axis with y.

%Finally gives the title "Multiple Graphs"

```
clc
```

```
clear all
```

```
close all
```

```
X=0:0.1:8;
```

```
plot(X,exp(-X))
```

```
hold on
```

```
plot(X,sin(X))
```

```
hold off
```

```
xlabel('x')
```

```
ylabel('y')
```

```
title('Multiple Graphs')
```

```
grid
```

L04_Q10: Plot the parametric graph of the circle with center at (0; 0), radius 1, and with increment of 0:01 in the domain set. Label the horizontal axis with x and vertical axis with y. Make the shape of the graph square. Keeps the title of the graph "Parametric plot of the circle". (P#98-100 - LECT#04)

%Plot the parametric graph of the circle with center at (0; 0),

% with radius 1 and with increment of 0:01 in the domain set.

```
clc
```

```
clear all
```

```
close all
```

```
t=0:0.01:1;
```

```
plot(cos(2*pi*t),sin(2*pi*t))
```

```
xlabel('x')
```

```
ylabel('y')
```

```
title('Parametric Plot of the Circle')
```

```
grid
```

L04_Q11: Plot the contour plots of the expression $x^2 + y^2$ using meshgrid and contour commands. Label the horizontal axis with x and vertical axis with y. Make the shape of the graph square. Keeps the title of the graph "Contour plotting of squares". (P#102-104 - LECT#04)

%Plot the contour plots of the expression $x^2 + y^2$ using meshgrid and contour commands,

%Label the horizontal axis with x and vertical axis with y.

%Make the shape of the graph square.

%Keeps the title of the graph "Contour plotting of squares".

```
clc
```

```
clear all
```

```
close all
```

```
[X,Y]=meshgrid(-2:0.1:2,-2:0.1:2);
```

```
contour(X,Y,X.^2+Y.^2)
```

```
axis square
```

```
xlabel('x')
```

```
ylabel('y')
```

```
title('Contour Plotting of Squares')
```

**Important and Solved Questions for Mid Term Exam**

```
syms x
ezplot('sin(x)+cos(x)',[0,4])
hold on
ezplot('exp(-x)+x^2',[0,4])
xlabel('x')
ylabel('y')
axis square
title('The Graph of Multiple curves using ezplot command')
grid
```

L04_Q15: Plot the contour plots of the circles x^2+y^2 of radius $1, 2, \sqrt{2}, \sqrt{3}$. Label the horizontal axis with x and vertical axis with y. Make the shapes of the graph square. Keeps the title of the graph "Contour Plots of Circles". Keeps the layout of the graph grid.

(P#109 - EXERCISE - LECT#04)

%Plot the contour plots of the circles x^2+y^2 of radius 1,2,72,73

%Label the horizontal axis with x and vertical axis with y.

%Make the shapes of the graph square.

%Keeps the title of the graph "Contour Plots of Circles".

%Keeps the layout of the graph grid.

clc

clear all

close all

syms a b

[X,Y]=meshgrid(-2:0.1:2,-2:0.1:2);

contour(X,Y,X.^2+Y.^2-1);

contour(X,Y,X.^2+Y.^2-2);

contour(X,Y,X.^2+Y.^2-sqrt(2));

contour(X,Y,X.^2+Y.^2-sqrt(3));

axis square

xlabel('x-axis')

ylabel('y-axis')

title('Contour Plots of Circles')

grid

L04_Q16: Plot the graph of $\sin(x)+\cos(x)$ and $e^{-x}+x^2$ on the interval $[0, 4]$ using plot, hold on and without use of hold on and hold off commands. Label the horizontal axis with x and vertical axis with y. Finally gives the title "The Graph of Multiple Curves using plot command". (P#108 - EXERCISE - LECT#04)

%Plot the graph of $\sin(x)+\cos(x)$ and $\exp(-x)+x^2$

%on the interval $[0, 4]$ using plot command and without use of hold on and hold off commands.

%Label the horizontal axis with x and vertical axis with y.

%Finally gives the title "The Graph of Multiple Curves using plot command".

%Keeps the layout of the graph as grid.

clc

clear all

close all

x=0:0.1:4;

f1=sin(x)+cos(x);

f2=exp(-x)+x.^2

plot(x,f1,x,f2)

xlabel('x')

**MTH643 - INTRODUCTION TO MATLAB****From: ABU SULTAN****Page 1 of 22****Important and Solved Questions for Mid Term Exam**

L02_Q01: Find value of following trigonometric function at $x = \frac{\pi}{2}$ using subs command

Cos(x), Sin(x), Tan(x).

(FOLLOW P#37-38 - LECT#02)

```
clear all
close all
clc
syms x a
a=cos(x)
subs(a,x,sym('pi/2'))
a=sin(x)
subs(a,x,sym('pi/2'))
a=tan(x)
subs(a,x,sym('pi/2'))
```

L02_Q02: Find value of following trigonometric function at $x = \frac{\pi}{2}$ using sym command

Cos(x), Sin(x), Tan(x). (FOLLOW P#37-38 - LECT#02)

```
close all
clear all
clc
cos(sym(pi/2))
sin(sym(pi/2))
tan(sym(pi/2))
```

L02_Q03: Find the value of the following using subs command. $u(x,y) = 2x + xy^2 + 3y$

at $x=5$, $f(x) = \sin x$ at $x = \frac{\pi}{2}$

(Follow P#036-42 - LECT#02)

(ABU SULTAN MID TERM Q#3(a) - 26 DEC 2019 - 12:00 pm)

```
clear all
close all
clc
syms x y u f
u=2*x+x*y^2+3*y;
subs(u,x,5)
f=sin(x)
subs(f,x,pi/2)
```

L02_Q04: Find the value of the following polynomials at $x=2$ using subs command.

$p(x,y) = x^2 + 3y^2x + 10$, $u(x) = x^3 + 4x + 10$ (Follow P#036-42 - LECT#02)

```
clear all
close all
clc
syms x y p u
p=x^2+3*y^2*x+10
subs(p,x,2)
u=x^3+4*x+10
subs(u,x,2)
```



MTH643 - INTRODUCTION TO MATLAB

From: ABU SULTAN

Page 2 of 22

Important and Solved Questions for Mid Term Exam

L03_Q05: Find the solution of $x^2 - y = 2$ and $y - 2x = 5$ (P#066-68 - LECT#03)

```
close all
clear all
clc
syms x y
[x,y]=solve('x^2-y=2','y-2*x=5')
x(1)
y(1)
```

L03_Q06: Find solution of equation $2x^3 + x^2 + 3x + 4 = 0$ (FOLLOW P#066-68 - LECT#03) (ABU SULTAN MID TERM Q#3(b) - 26 DEC 2019 - 12:00 pm)

```
close all
clear all
clc
syms x
solve('2*x^3+x^2+3*x+4=0')
```

L04_Q07: Solve by solve command. $2x + 7y = 3$, $3x^2 + 2y = 1$. Also plot graph for $3x^2 + 2y = 1$. Name it parabola, give x-label and y-label also.

(FOLLOW P#066-71 - LECT#03) and (FOLLOW P#82,86 - LECT#04)

```
clc
close all
clear all
syms x y
solve('2*x+7*y=3','3*x^2+2*y=1')
ezplot('3*x^2+2*y=1',[-2,2])
xlabel('x-axis')
ylabel('y-axis')
title('Parabola')
```

L04_Q08: Plot the graph of e^{-x} and $\sin(x)$ on the interval $[0, 8]$ using ezplot, hold on and hold off commands. Label the horizontal axis with x and vertical axis with y. Finally gives the title "Multiple Graphs". (P#90-92 - LECT#04)

```
%Plot the graph of e^(-x) and Sin(x)
%on the interval [0, 8] using ezplot, hold on and hold off commands.
%Label the horizontal axis with x and vertical axis with y.
%Finally gives the title "Multiple Graphs"
```

```
clc
clear all
close all
syms x
ezplot(exp(-x),[0,8])
hold on
ezplot(sin(x),[0,8])
hold off
xlabel('x')
ylabel('y')
title('Multiple Graphs')
grid
```

Screenshot has been saved to/Pictures/Screenshot